

Hunter Pettus

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EDUCATION

University of California, Santa Cruz

Santa Cruz, CA

Bachelor of Science in Computer Science, GPA: 3.90

Sep. 2023 – June 2027

Courses: System Design, Data Structures & Algorithms, C & Embedded Systems, Discrete Math, Linear Algebra, Machine Learning, Computer Architecture, Computer Graphics, & Artificial Intelligence

EXPERIENCE

AMD

Austin, TX

Software Engineer Intern

May 2026 – August 2026

- Developed a globally deployed inventory management platform built with React, TypeScript, Python, and SQL, serving 15+ directors, PMs, and analysts across 3 global AMD sites
- Architected a **multi-agent LLM system with tool-calling and RAG** over live inventory data, autonomously resolving hardware allocation requests and surfacing shortage alerts — reducing resolution time from **hours to minutes**
- Automated end-to-end inventory reconciliation and order lifecycle workflows across EPYC CPUs, memory modules, and server infrastructure — including approval routing, audit history, and stakeholder notifications — supporting capacity planning, shortage detection, and memory resource forecasting

Technical Program Manager Intern

Jan. 2026 – May 2026

- Coordinated execution of 3 EPYC Venice server CPU programs across 6 engineering, supply chain, and validation teams, accelerating milestone delivery by **2+ weeks** through **critical path tracking** and proactive blocker escalation
- Facilitated weekly cross-functional syncs, driving alignment on hardware readiness and resource dependencies; created dashboards to surface program status and milestone progress across the EPYC Venice server portfolio

Arthrex

Santa Barbara, CA

Software Engineer Intern

June 2025 – September 2025

- Rebuilt legacy healthcare case management system handling **100,000+ patient cases**, reducing query times from **10 seconds to milliseconds** and eliminating 25-second loading delays, enabling the entire customer service team to resolve patient complaints with near-instant data access
- Migrated from C# .NET Framework to Node.js **microservices architecture** with **RESTful API design**, leading cross-functional feedback sessions to redesign front-end workflow and deploy directly to production, achieving **80% test coverage** on 10k+ lines of code

PROJECTS

Drone Orchard Monitoring System | Python, HuggingFace, DJI Tello SDK, React, ROS, OpenCV

- Deployed LocateAnything-3B (**HuggingFace VLM**) on a physical DJI Tello drone pipeline to classify 9 fruit ripeness stages across multiple fruits at **90%+ accuracy**, with uncertainty quantification flagging ambiguous predictions within a 30% confidence differential
- Estimated per-fruit distance in real time using a calibrated pinhole camera model with ripeness-specific diameter lookup tables — deriving **3D spatial localization from 2D bounding boxes** without a depth sensor
- Architected end-to-end pipeline from aerial capture through **geometric deduplication**, Flask API, and React dashboard tracking **1,000+** detections across sessions with ripeness distribution analytics

Edge-Deployable Seismic Event Detector | Python, PyTorch, ONNX Runtime, SeisBench

- Built a seismic event detector for real-time, off-grid field deployment, no cloud or GPU required, by distilling a 377k-parameter EQTransformer into a **48k-parameter** model while achieving **F1 0.994**
- Quantized to **INT8 ONNX** for CPU-only inference at **1.2ms/window** ($2.7\times$ faster than PyTorch), validated for real-time streaming on low-power edge hardware like Raspberry Pi

TECHNICAL SKILLS

Languages: Java, Python, C/C++, C#, Typescript, HTML/CSS, Assembly, SQL

Developer Tools: Git, AWS, Linux, VS Code, Microsoft Office, Power BI, Unreal Engine, Angular, Docker, ROS, OpenCV

Libraries: pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch